

World Life Expectancy

Which countries gained the most life expectancy over 16 years, how strongly it tracks GDP, and how developed and developing nations compare.

Tools SQL. Data cleaning, window functions, aggregation

Data WHO life expectancy dataset. 2,938 country-year records, 183 countries, 2007 to 2022

The questions

- Which countries improved life expectancy the most, and which improved the least?
- Has average life expectancy actually risen worldwide over the period?
- How closely does life expectancy track GDP, and does the relationship hold at both ends?
- How large is the gap between developed and developing nations?
- Does average body mass in a population relate to life expectancy?

The data

One row per country per year, with life expectancy, GDP, adult mortality, BMI, immunisation rates and a developed or developing status flag. Three issues had to be resolved in SQL before the analysis queries could run.

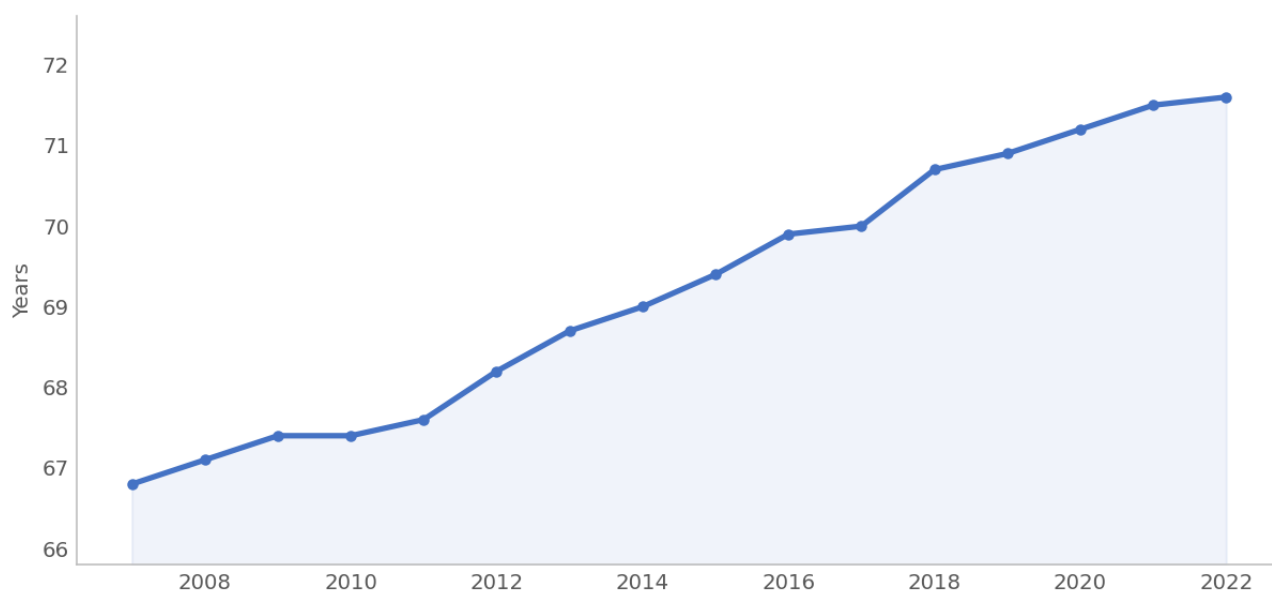
Problem	What I did
3 duplicate country-year rows	Used ROW_NUMBER() partitioned by country and year to identify and delete the extras
8 rows with a blank development status	Filled each from the status recorded for that country in other years, via a self-join
12 rows with a missing or zero life expectancy value	Filled by averaging the same country's value in the year before and the year after, via a self-join on Year-1 and Year+1

2,928 rows remained after cleaning. The window-function dedup and the neighbour-year fill are the two techniques worth pointing to directly in the SQL file.

Life expectancy is rising almost everywhere

The global average climbed from 66.8 years in 2007 to 71.6 in 2022, a gain of 4.8 years over the period, with no year-over-year decline anywhere in the series.

Global Average Life Expectancy, 2007 to 2022

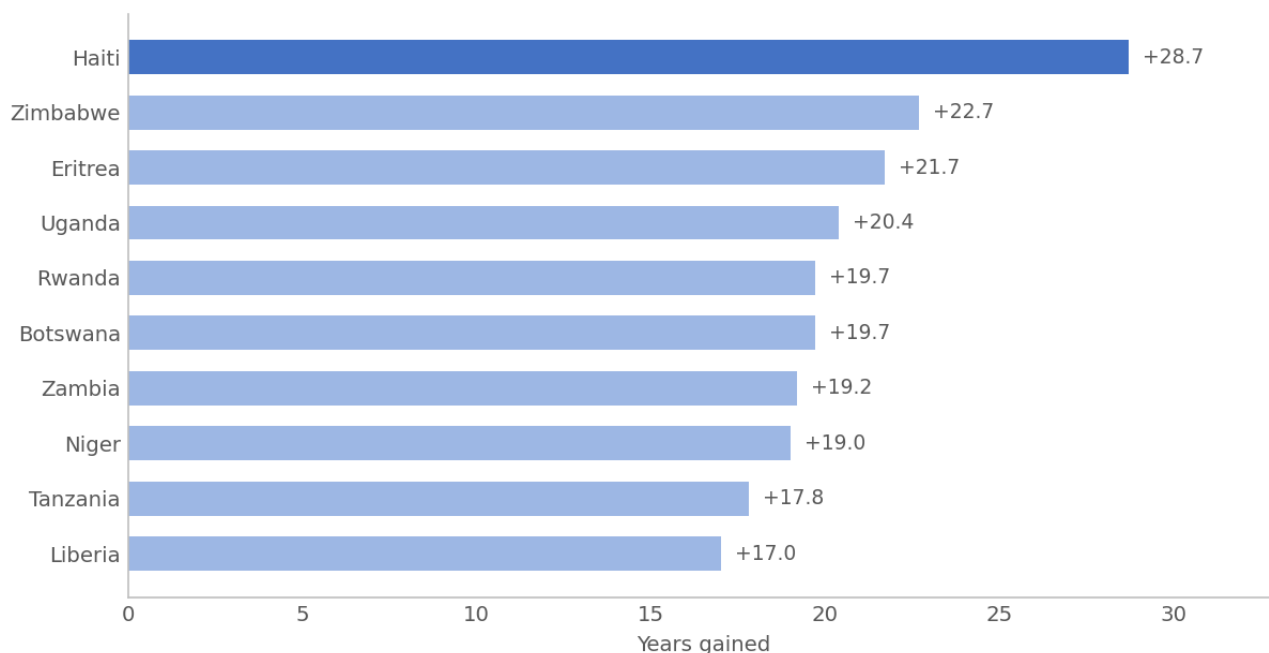


Global average life expectancy across all countries, 2007 to 2022.

The gains are concentrated in countries that started lowest

Haiti gained 28.7 years, the largest increase in the dataset, followed by Zimbabwe, Eritrea, Uganda and Rwanda, each gaining close to 20 years. Every country in the top ten started below 51 years in its worst recorded year. Countries that started already above 65 years, like Guyana and Kuwait, gained under 2 years over the same period, simply because there was less room to close.

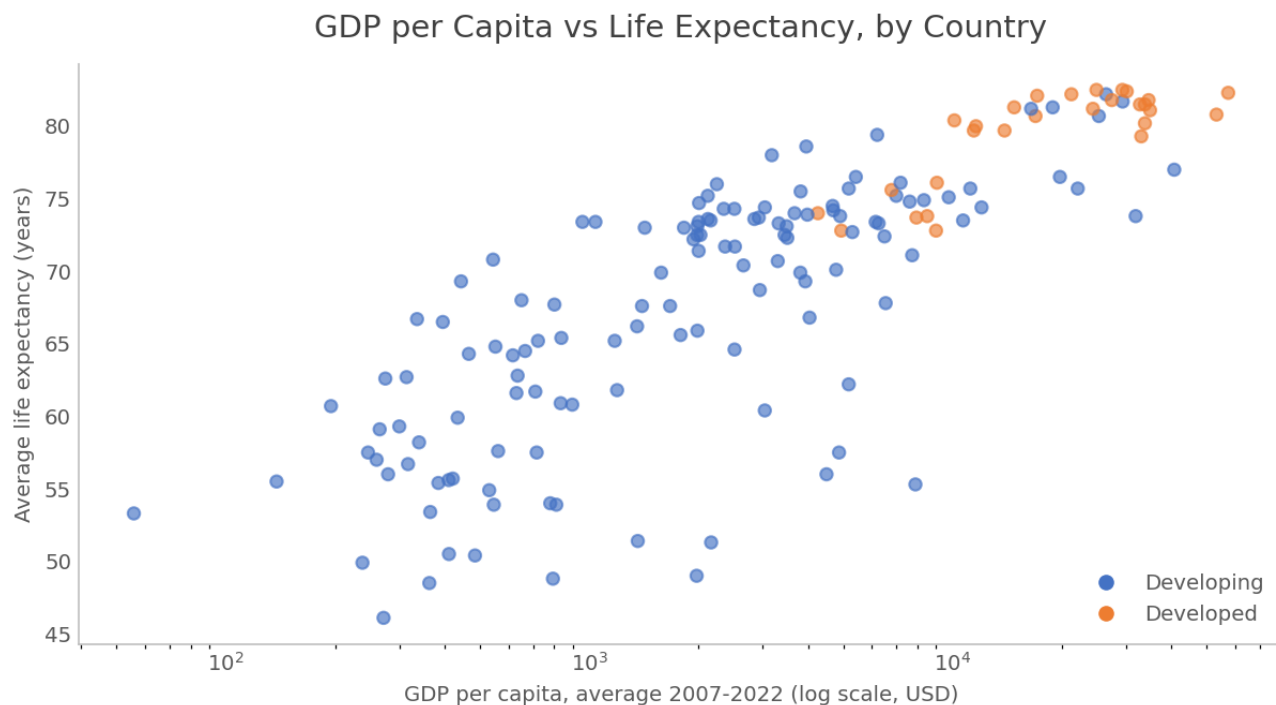
Largest Life Expectancy Gains, 2007 to 2022



The ten largest gains in life expectancy, minimum to maximum year recorded, 2007 to 2022.

GDP tracks life expectancy closely, with diminishing returns

Averaging each country across the full period, GDP per capita and life expectancy correlate at 0.61. Splitting the dataset at \$1,500 GDP per capita, countries above that line average 74.4 years against 65.0 below it, a 9.4 year gap. The relationship is strongest at the low end: moving from extreme poverty to lower-middle income buys several years of life expectancy. Past roughly \$10,000 per capita the curve flattens, and further GDP growth buys comparatively little.

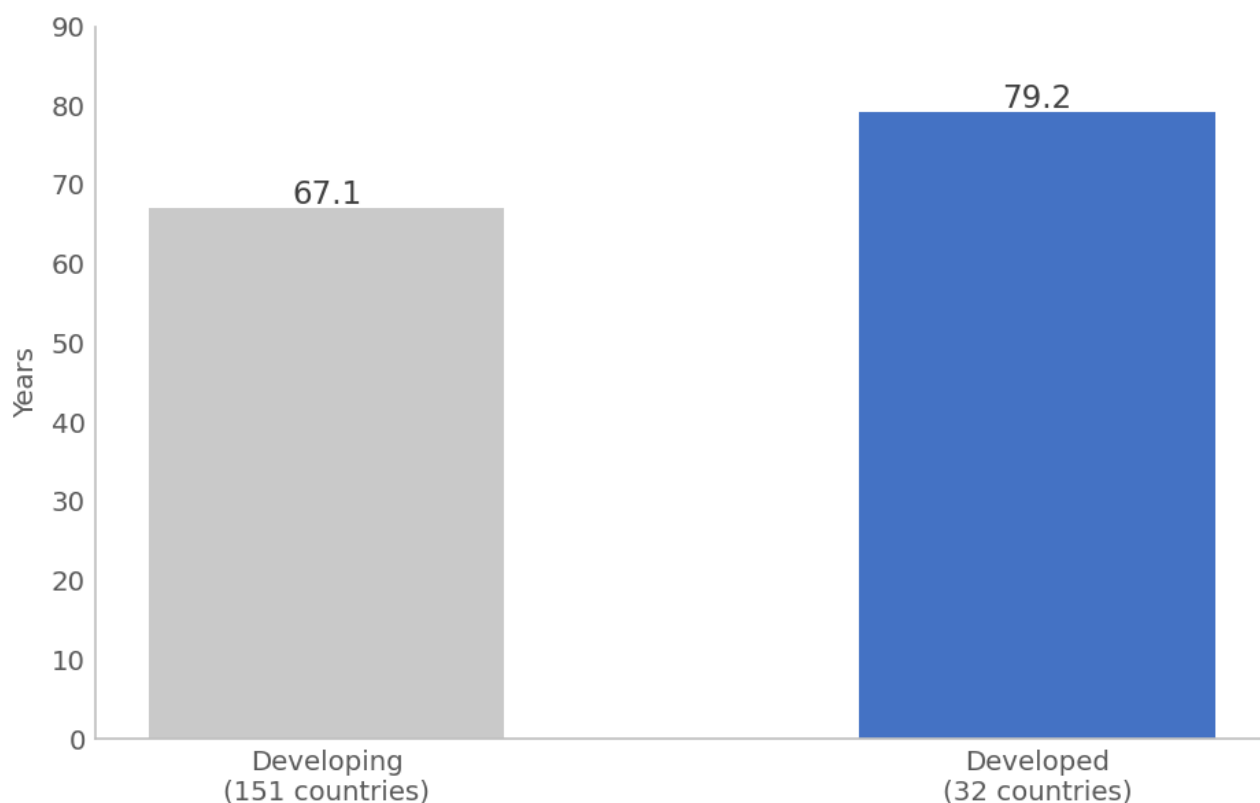


Average GDP per capita against average life expectancy, one point per country. Note the log scale on GDP.

The development-status gap is large and structural

The 32 countries flagged as developed average 79.2 years against 67.1 for the 151 flagged as developing, a 12.1 year gap. That is wider than the raw GDP split above, which suggests status captures more than income alone, likely healthcare infrastructure and public health systems that GDP per capita does not fully reflect.

Average Life Expectancy by Development Status



Average life expectancy, developed versus developing countries.

Interpretation and limitations

The practical read is that early GDP growth in low-income countries has an outsized health payoff, consistent with the top-gains list being dominated by countries that started poorest. Beyond that income range, the marginal effect of GDP is small, so a country already past it looking to raise life expectancy further should look at levers other than pure income growth.

This is observed correlation, not causation. GDP, healthcare access and public health policy move together, and the dataset cannot separate which one is doing the work. The status split in particular is a proxy: it groups countries by a classification that already reflects policy and infrastructure differences a GDP number alone would not capture.

What is in the file

Section	Contents
Data cleaning	Duplicate detection and removal, status backfill, life expectancy interpolation
Data analysis	Six queries: country-level gains, yearly trend, GDP by country, high vs low GDP buckets, status comparison, BMI by country, plus a rolling mortality total using a window function

Portfolio project built on a public WHO life expectancy dataset.